

Complishon: a Modular engine and its extensions

<http://www.pharo.org>



ESUG 2025



Outline

- Challenges
- Complishon in a nutshell
- Dependent packages
- History and user selection
- [Preview] Tolerance

Code Completion

- Our best friend
 - A love-and-hate affair!
 - Well, the syntax can be ambiguous
 - Your flow of thoughts is in your head :)
-
- Often not enough context
 - < 110 ms

We want everything and the inverse

- Often our flow is not monotone
- Why does the completion not do X or Y?

Complishon

- Originally developed by G. Polito
- Maintained and enhanced by S. Ducasse and O. AbedelKader

Keyword challenge!

- “1 between: 2 an”

Syntax of partially specified messages is ambiguous!

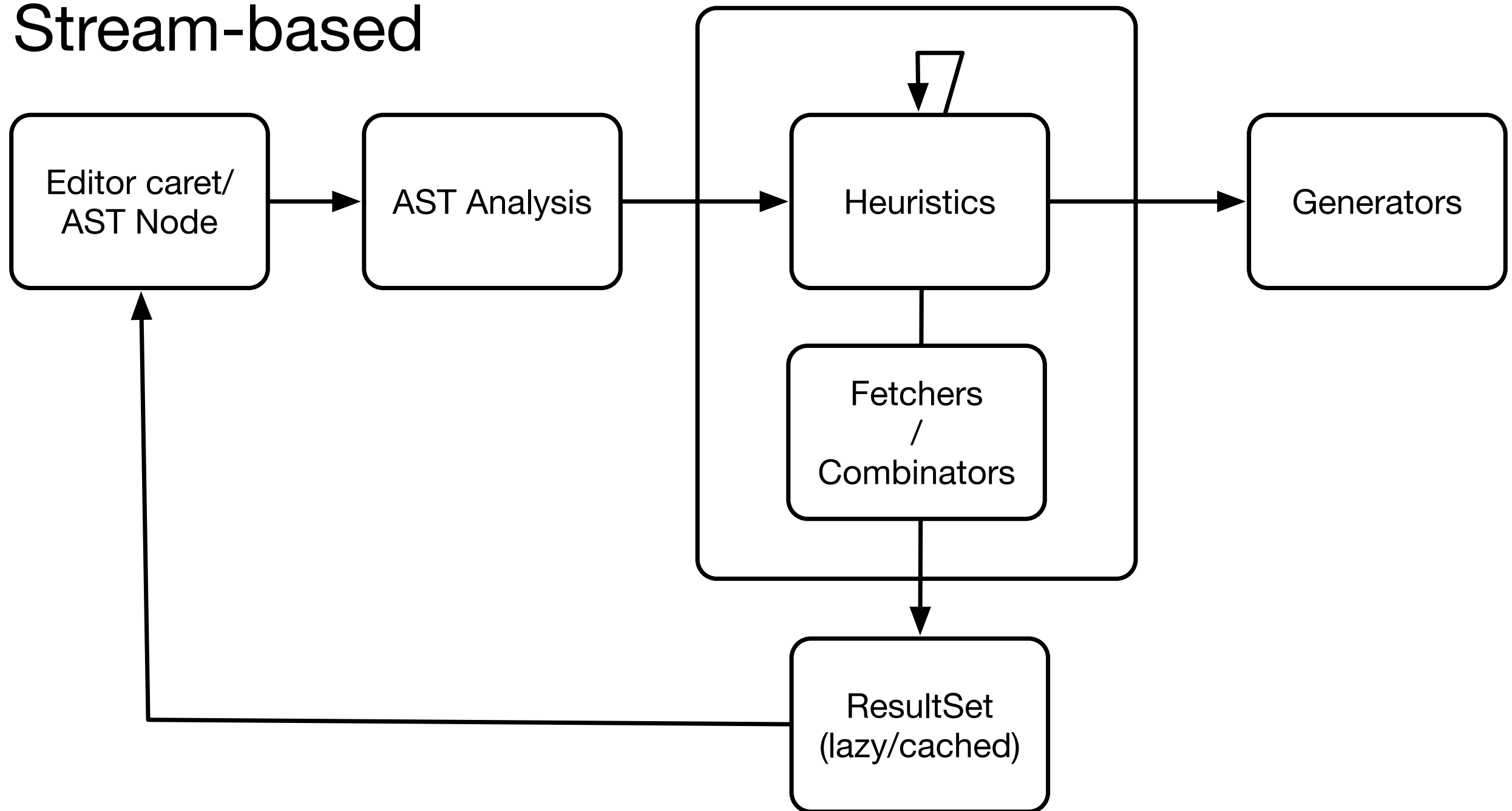
- 1 between: 2 anElephant
- 1 between: 2 and: 10

Global challenges

- Global names are not sorted
- Selectors are not sorted

101 on Complishion architecture

Stream-based

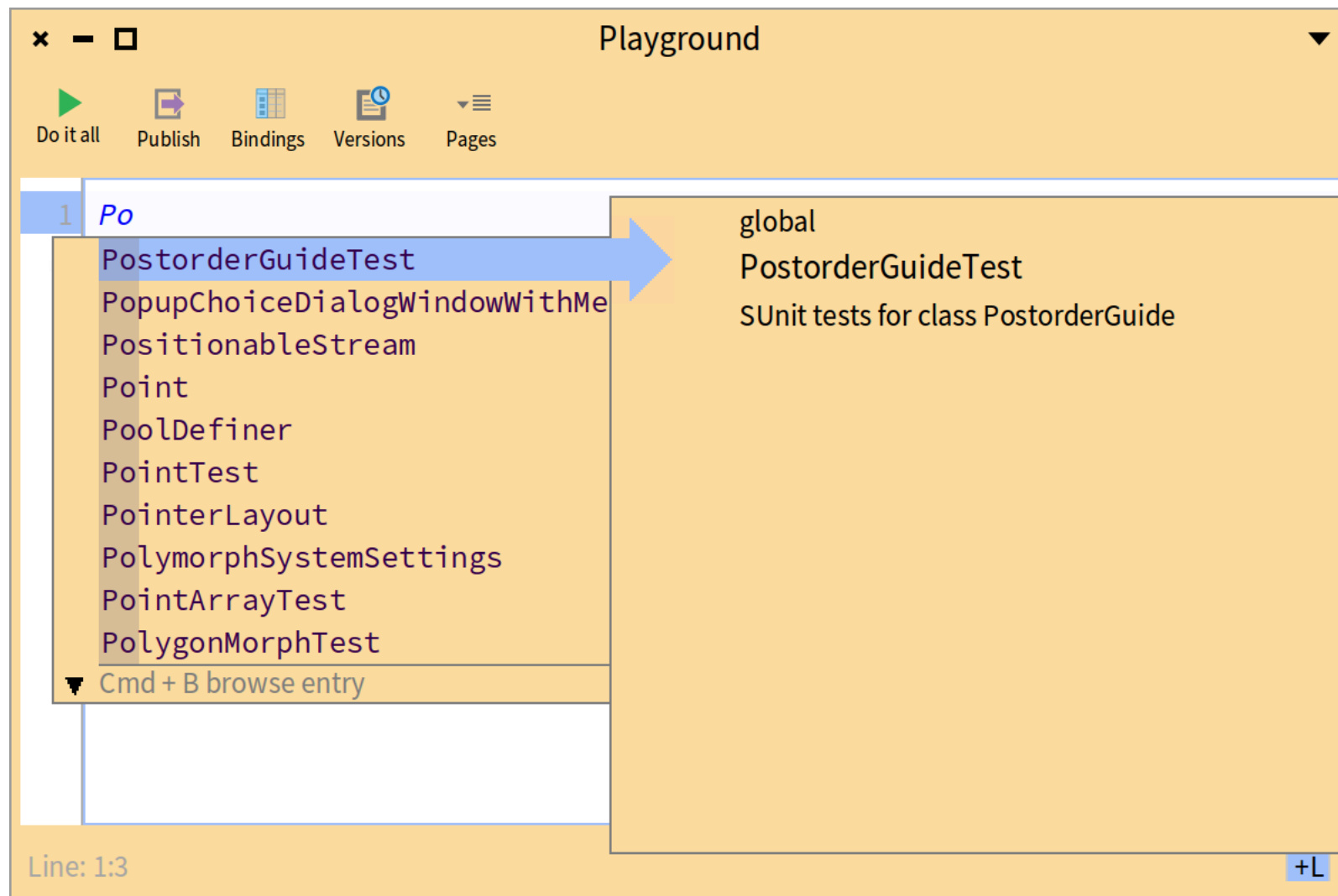


101 on Complishion

- Lazy fetchers grouped in heuristics
- Attached/triggered to/by a semantic model of the code
- Lazy means = 20 by 20
- Do not scan the full system
- Handle backspace, duplicated entries
- Fast!

Example: Global lookup

Classes are not sorted in global namespace



Example: instance var

- Inheritance

The screenshot displays the Morph IDE interface. The top pane shows the class hierarchy with **BorderedMorph** selected. The right pane shows the **instance side** of the **borderColor** method, with **overrides** highlighted. The bottom pane shows the **borderColor2** method, with a red exclamation mark icon and a blue arrow pointing to the **borderWidth** instance variable. A tooltip for **borderWidth** is visible, showing the text **Cmd + B browse entry**. The bottom status bar indicates the current view is **Inst. side methc**.

Method: BorderedMorph>>borderColor

Kernel
Manifest
Support
Worlds
Extensions
Morphic-Deprecated
Morphic-Testing-Support
Morphic-Tests
Morphic-Widgets-Basic
Morphic-Widgets-ColorPicker
Morphic-Widgets-Extra
Morphic-Widgets-FastTable
Morphic-Widgets-FastTable-Test
Morphic-Widgets-Menubar

Morph
BorderedMorph
HandMorph
MorphExtension
MorphicRenderLoop
ScalingCanvasSavedPatch

instance side
extensions
accessing
accessing - defaults
geometry
initialization
menu
private
private - initialization
testing
overrides

acquireBorderWidth:
addBorderStyleMenuItems:hand:
addCornerGrips
addPaneSplitters
addPaneVSplitterBetween:and:
basicBorderColor:
basicBorderWidth:
borderColor
borderColor:
borderInset
borderRaised
borderStyle
borderStyle:
borderWidth
borderWidth:
changeBorderColor:

All Packages | Scoped View | Projects | Flat | Hier. | Inst. side | Class side | Methods | Vars | Class refs. | Implementors | Send

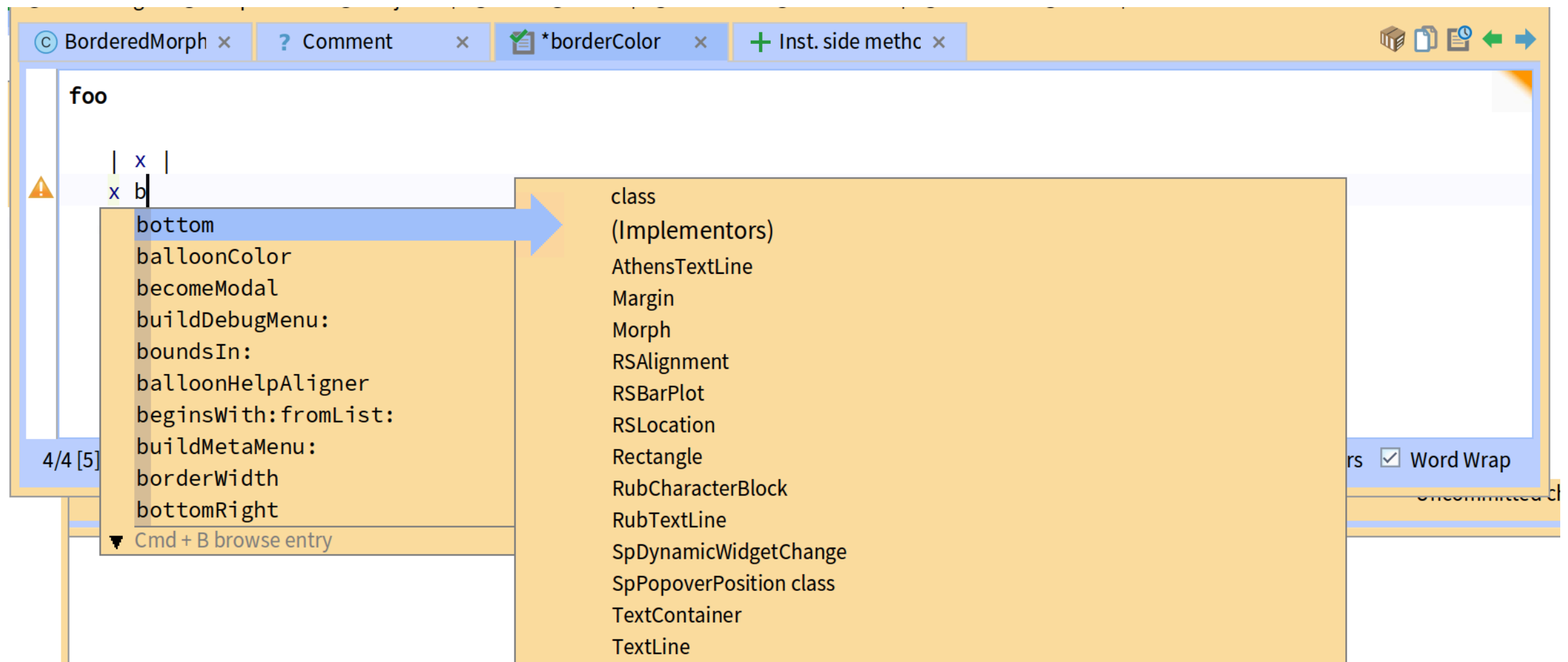
BorderedMorph x | Comment x | *borderColor x | + Inst. side methc x

borderColor2
! ^ bo
borderWidth
borderColor
bounds
Cmd + B browse entry

instance variable
(unknown)
-

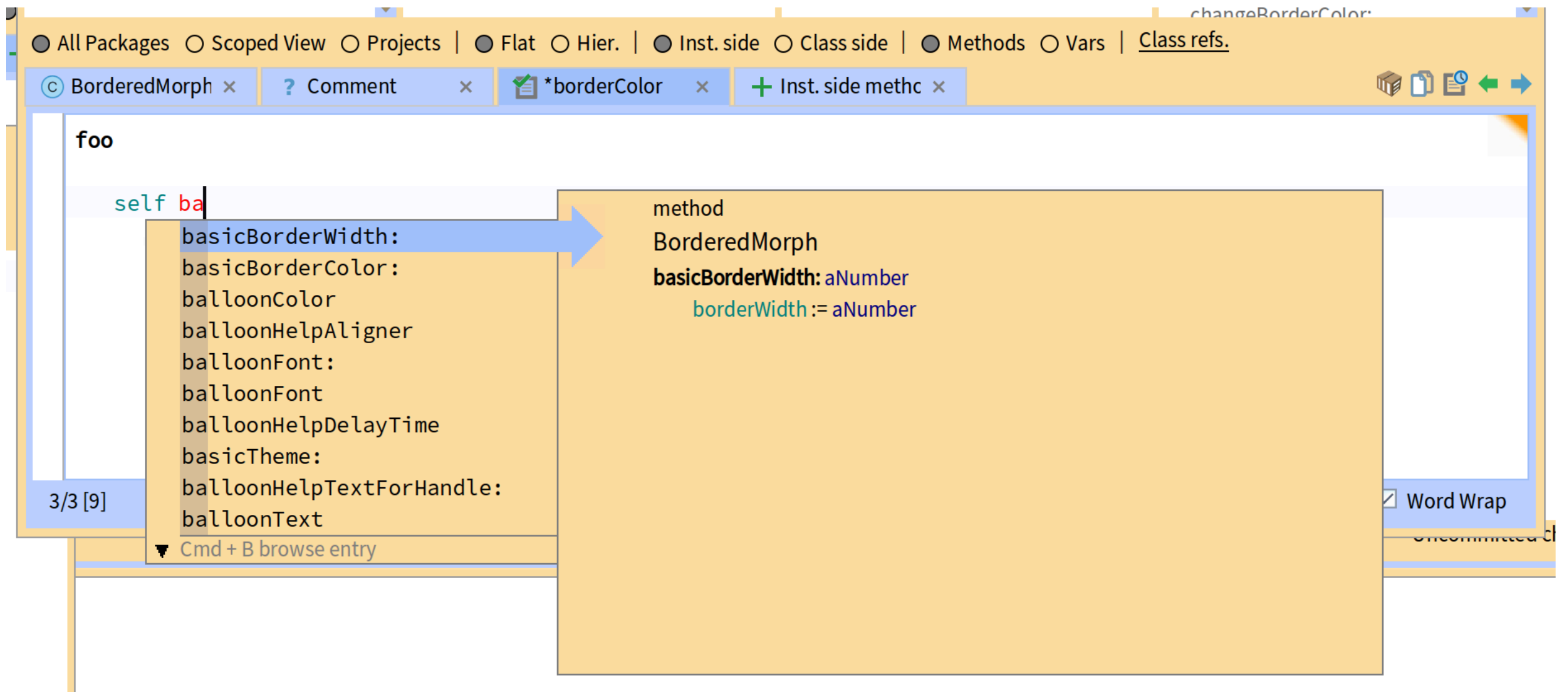
Example: unknown receiver

Lookup in global unsorted selector table



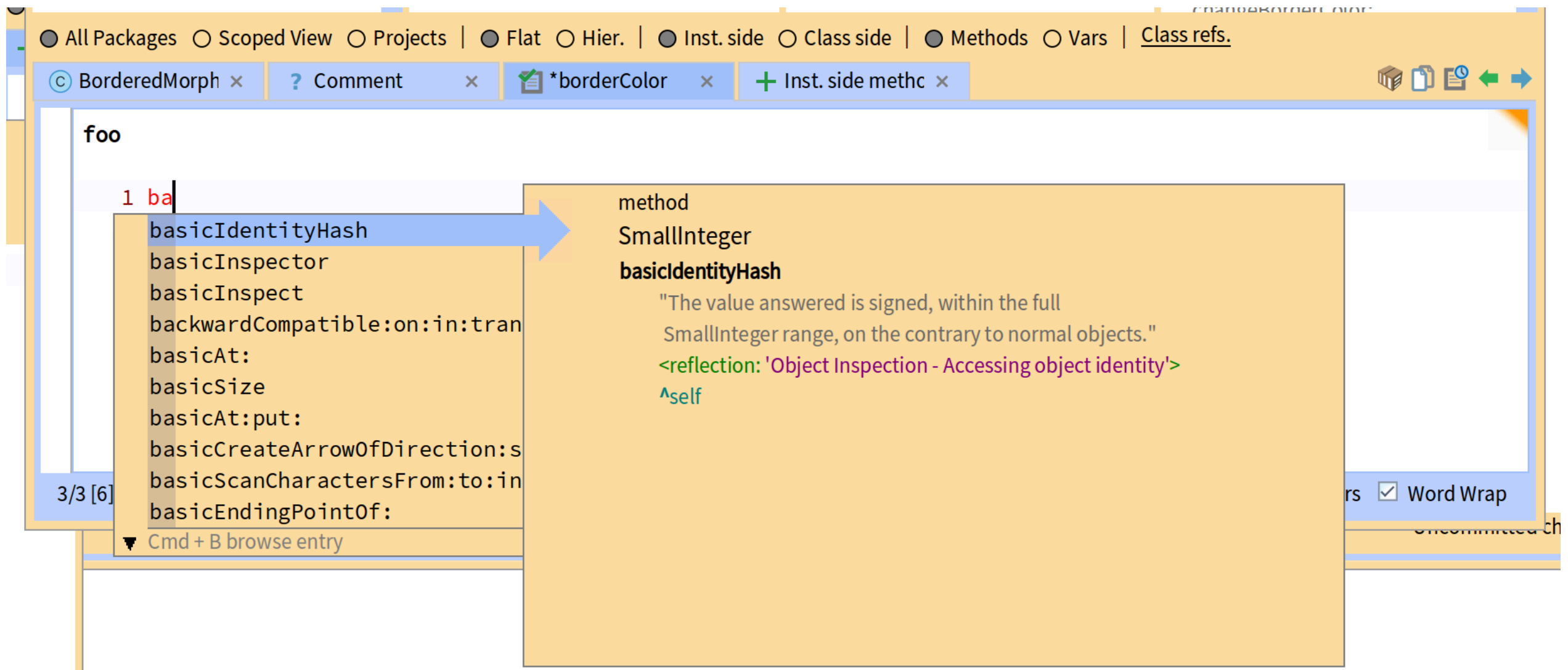
Example: self

In BorderedMorph: only methods and superclass



Example: known receiver

Lookup in SmallInteger/Number/Object



Example: known receiver

The screenshot shows an IDE window with a tab bar at the top containing: `BorderedMorph`, `Comment`, `*borderColor`, and `Inst. side methc`. The main editor area is titled `foo`. On the left, a list of methods is displayed, with `1 adaptToFraction:andSend:` highlighted. A blue arrow points from this entry to a larger, detailed view of the method on the right.

Method List (Left):

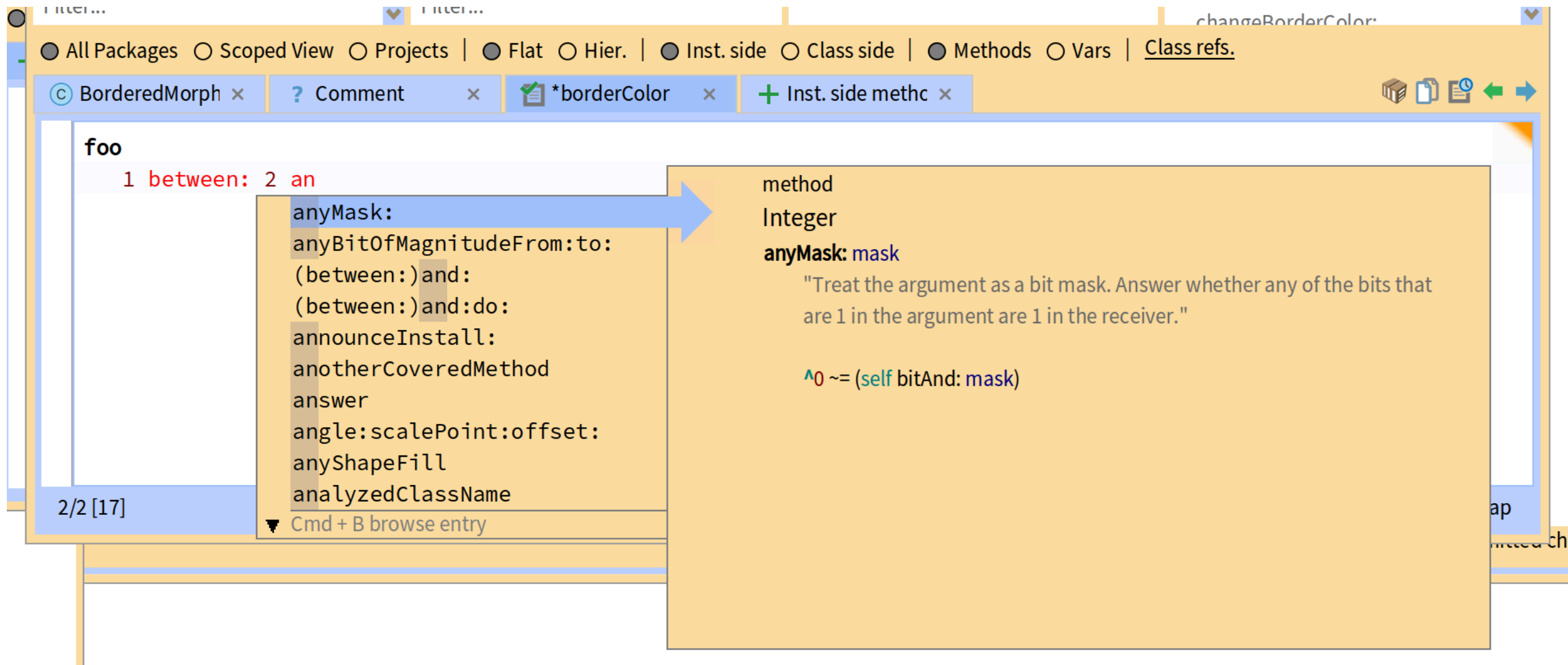
- `1 adaptToFraction:andSend:`
- `addAssignToFloatArray:`
- `adaptToInteger:andSend:`
- `adaptToFloat:andCompare:`
- `adaptToPoint:andSend:`
- `adaptToCollection:andSend:`
- `adaptToFloat:andSend:`
- `addModelYellowButtonMenuItems`
- `adaptToFraction:andCompare:`
- `addModelItemsToWindowMenu:`

Method Detail (Right):

method
Integer
adaptToFraction:rcvr andSend:selector
"If I am involved in arithmetic with a Fraction, convert me to a Fraction."
^rcvr perform: selector with: (Fraction numerator: self denominator: 1)

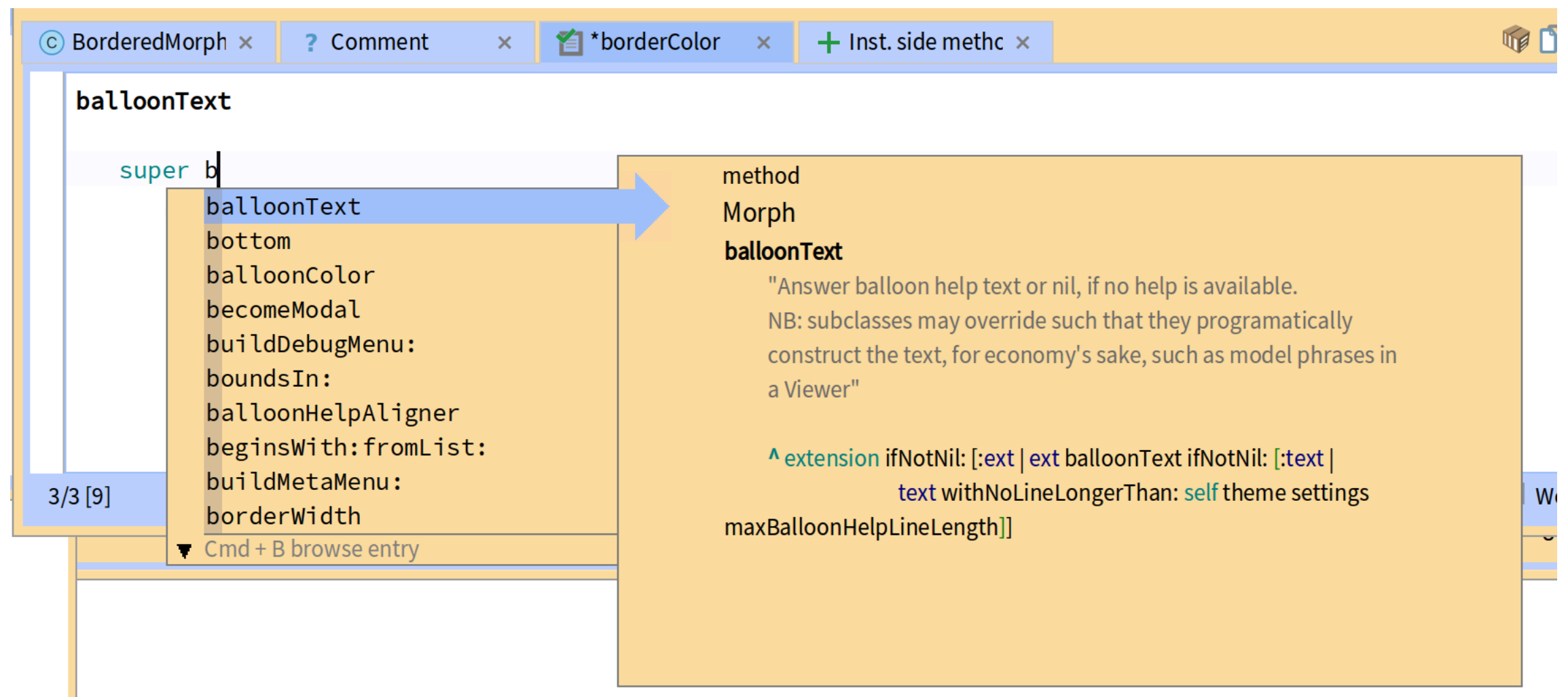
At the bottom left of the editor, there is a status bar with the text `3/3 [6]` and a tooltip that says `Cmd + B browse entry`.

Example: keywords



Example: super!

Notice the single character!



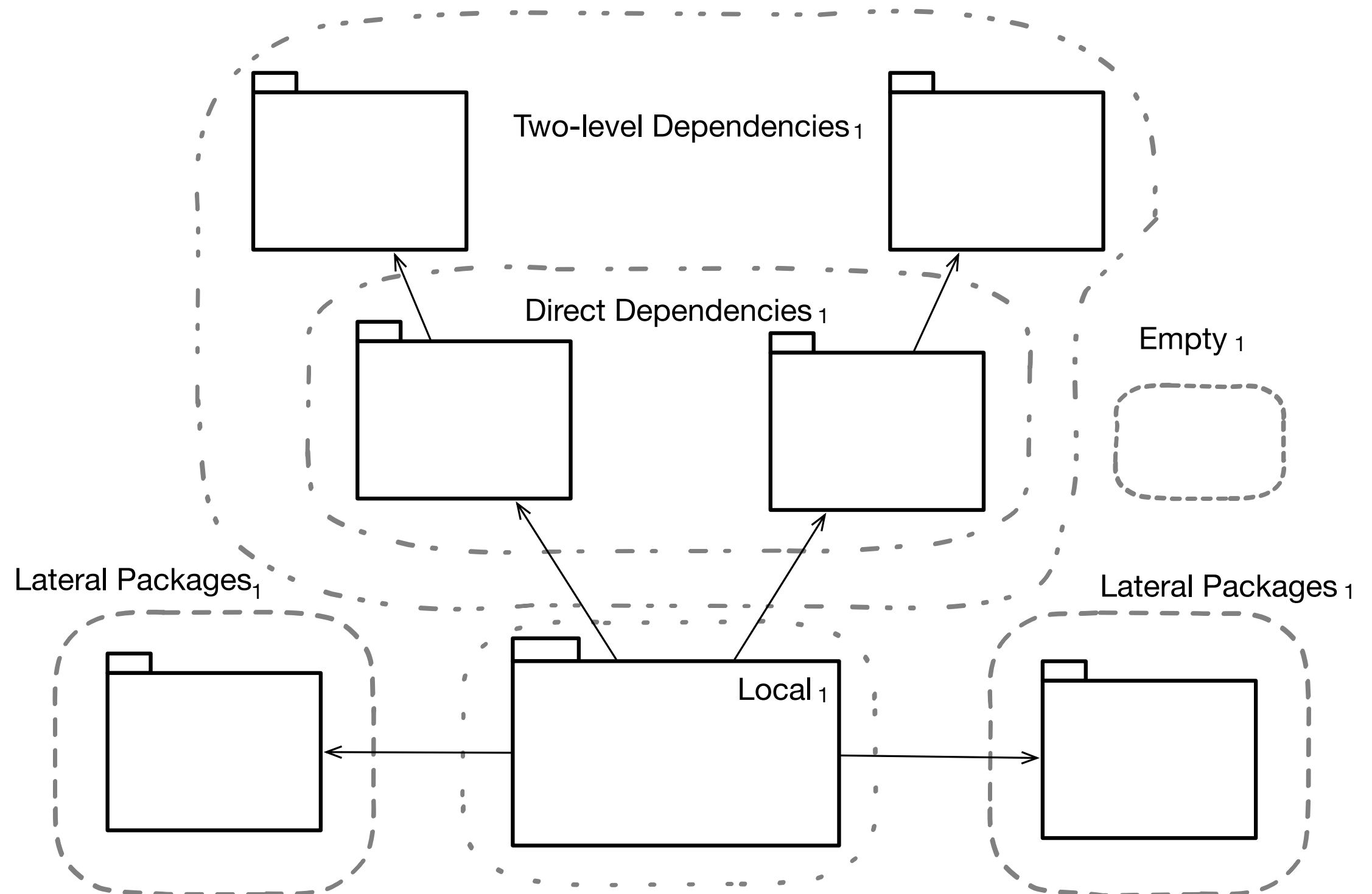
Customizable

- Special fetcher for Phausto only looks for certain tagged selectors!
- Easy to add your own

Package Dependency

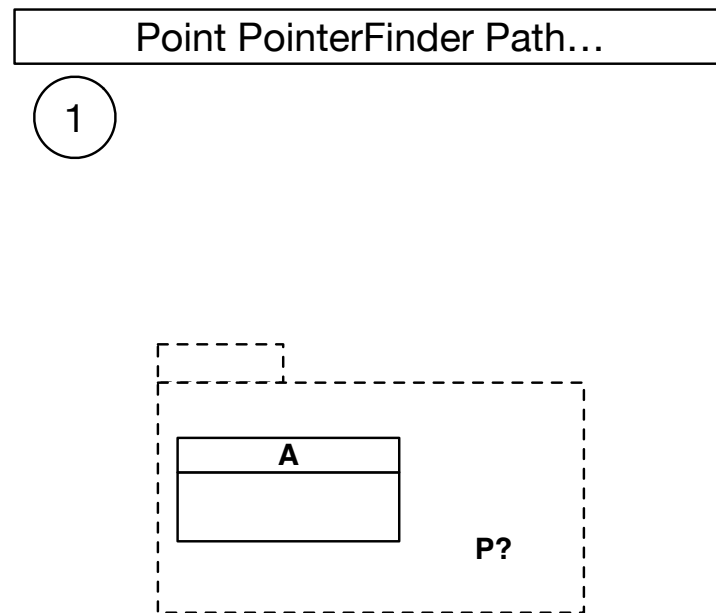
Can package dependencies improve
the completion?

Units of comparison

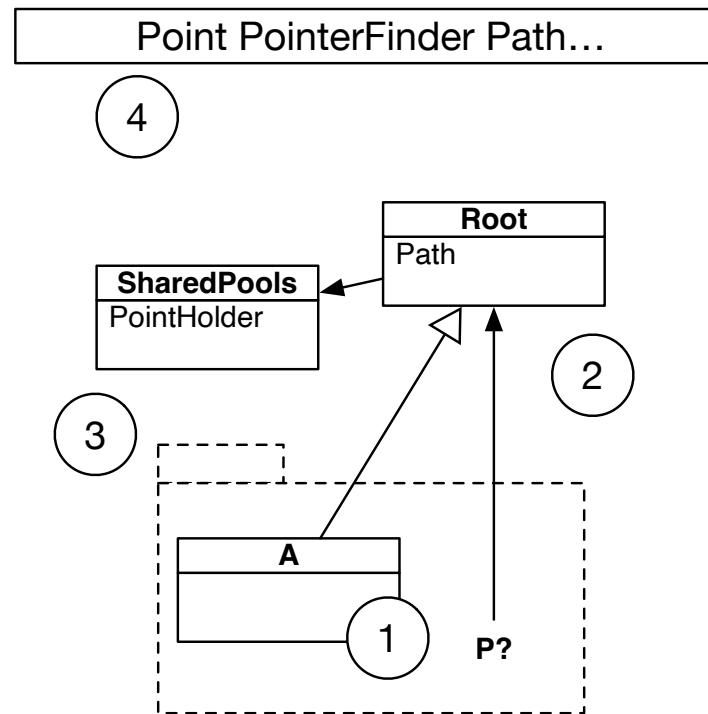


Scenarios

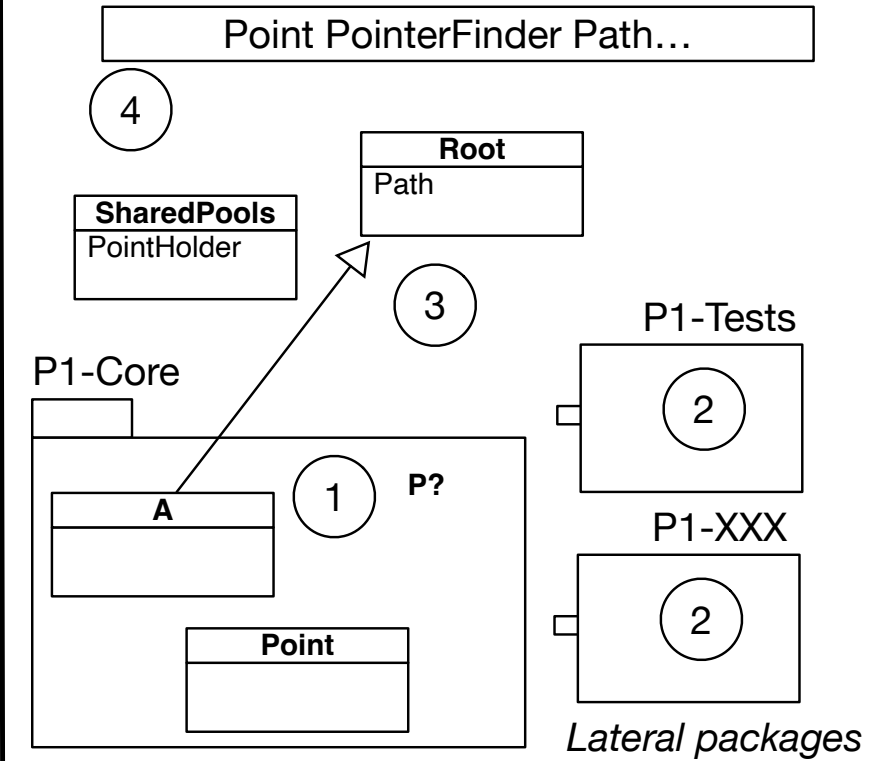
Baseline



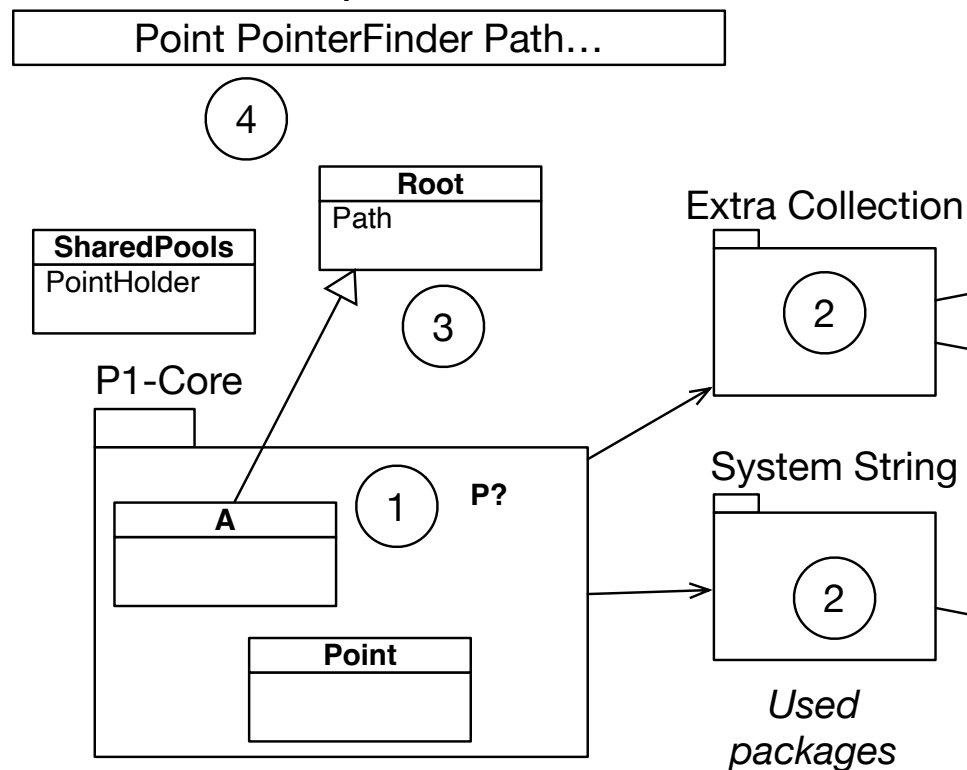
Semantics



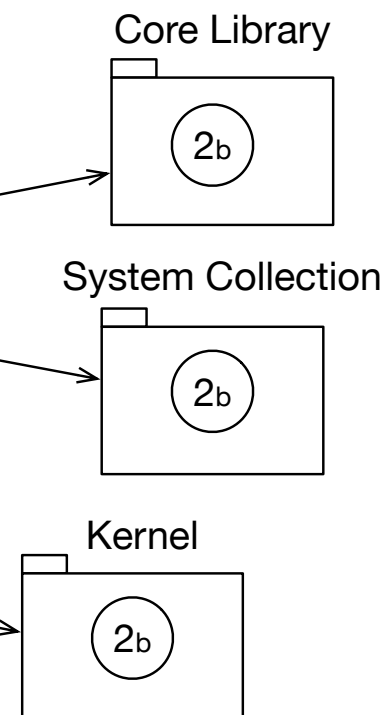
Laterals



Direct dependencies



2 Level dependencies



Evaluation overview

- Take all the messages of methods
- Change selectors to 2,3,4, .., 8 characters
- Compare with original selector

```
printOn: str
```

```
super printOn: str.
```

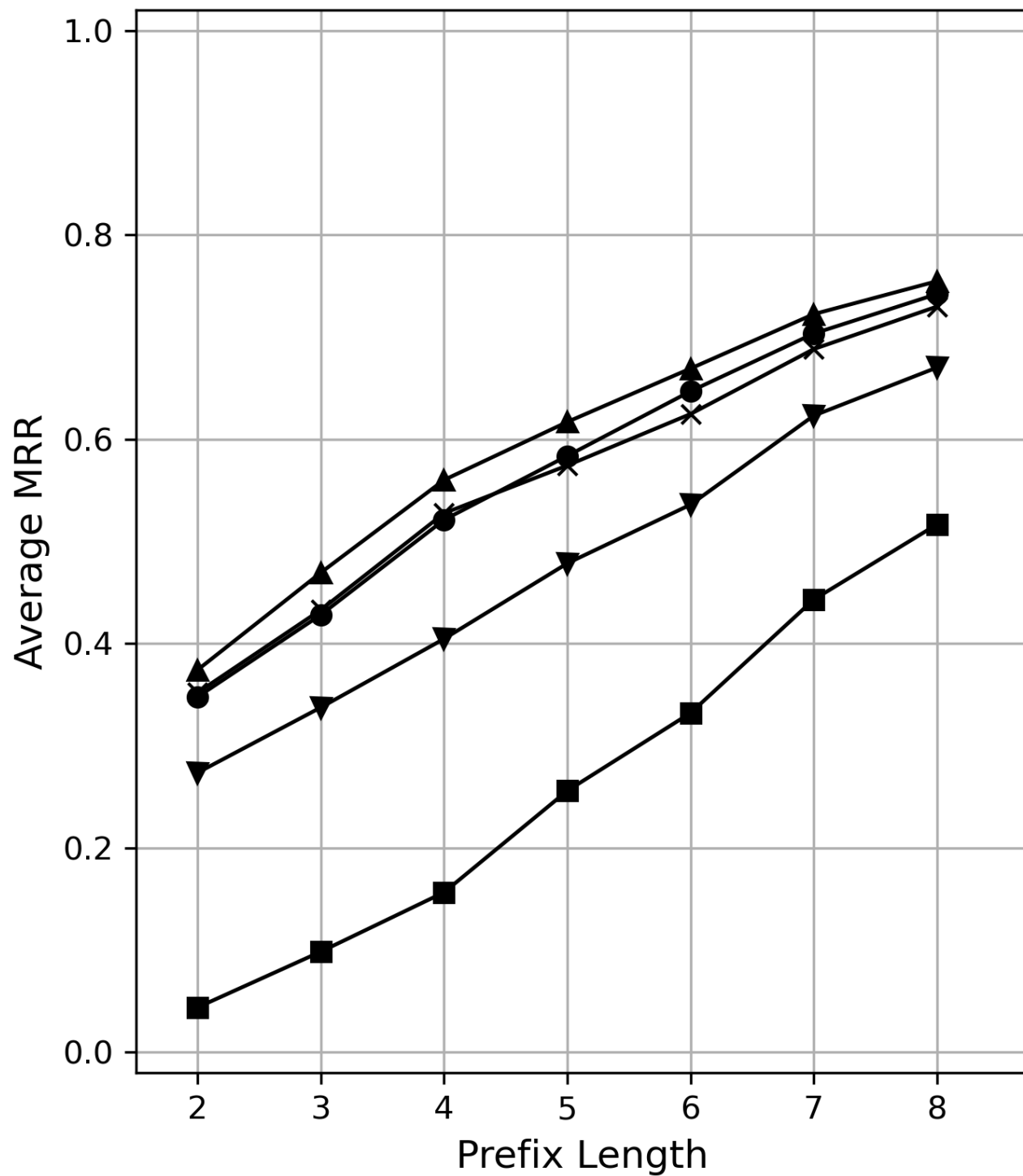
```
self printDeclarationOn: str
```

```
printOn: str
```

```
super prXXX str.
```

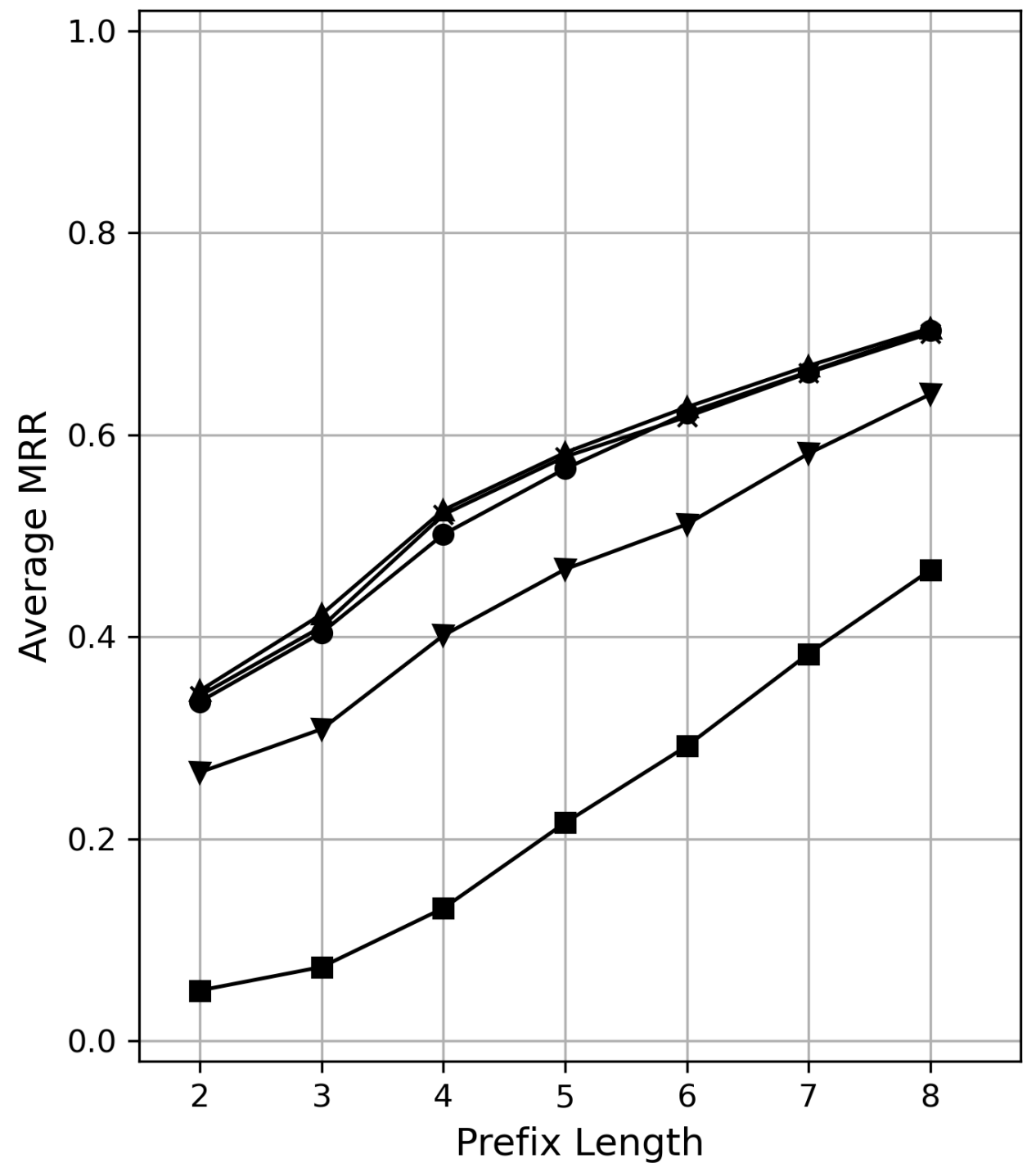
```
self prXX: str
```


Package Dependency Level



Strategy

- complete2LevelDependency
- completeBaseline
- ▲ completeDependency
- ✕ completeLateralPackage
- ▼ completeSemantics



Strategy

- complete2LevelDependency
- completeBaseline
- ▲ completeDependency
- ✕ completeLateralPackage
- ▼ completeSemantics

Package Dependency

- Better than no package dependency
- By default in Pharo

Complishion on steriods

- Package dependencies to better choices
- Coding history mixed with your last selections :)
- [preview] “Typpo tolerance”

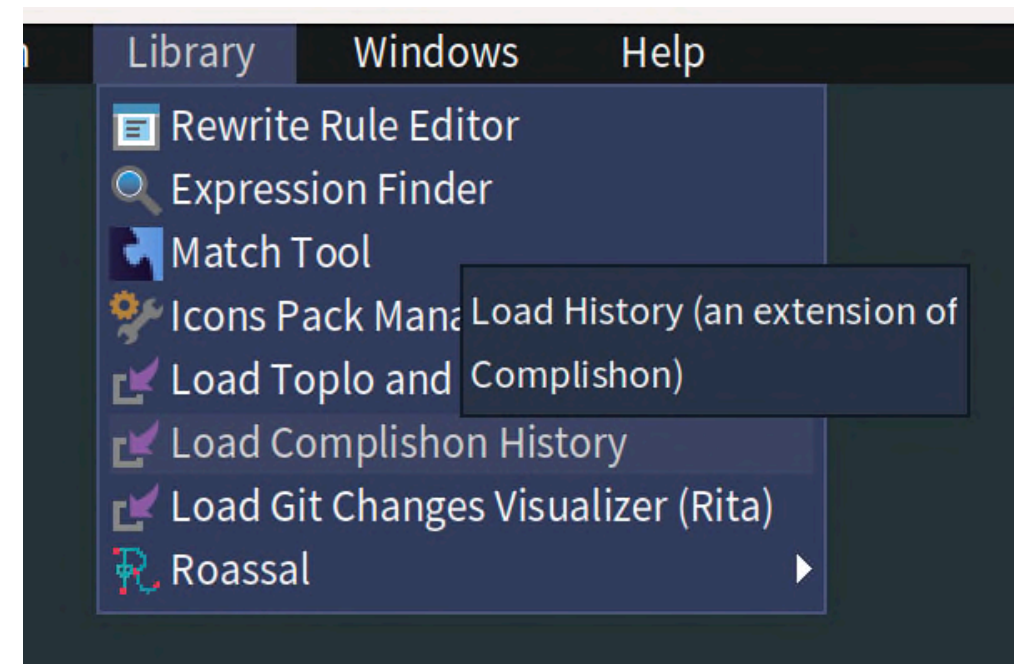
Using past activities

Optional but super cool

Take your **recent class/method addition/removal**

Take your **recent selections** in the menu

Use a recently use window



Recent edits

Take your **recent class/method addition/removal**

Multiple strategies

- More recent: more important $x \text{ t1} < y \text{ t2}$
- Statistics: number of $x >$ number of $x - 1$

Refill the fact base from GH commits

[Preview]

Refill the fact base

- using your recent commits
- granularity of order changes is the commit

But we can do better

You selected a class / method

You will select it again soon

Learn from your recent selection

The screenshot shows a 'Playground' window with a toolbar at the top containing icons for 'Do it all', 'Publish', 'Bindings', 'Versions', and 'Pages'. The code editor area has a line of code: `1 Ored`. A dropdown menu is open below the code, listing several classes: `OrderedCollection`, `OrderedCollectionTest`, `OrderedDictionary` (highlighted with a blue arrow), `OrderedDictionaryTest`, `OrderedIdentityDictionary`, and `OrderedIdentityDictionaryTest`. Below the list is the text 'Cmd + B browse entry'. To the right of the code editor is a documentation panel for the `OrderedDictionary` class. The panel has a title bar 'global OrderedDictionary' and contains the following text:

I am a collection that act as a Dictionary except that I use key insertion order when enumerating, printing, or returing collections of keys/values/associations, but not when testing for equality (but it does not matters in this case). I will assume that you know the Dictionary class in this comment.

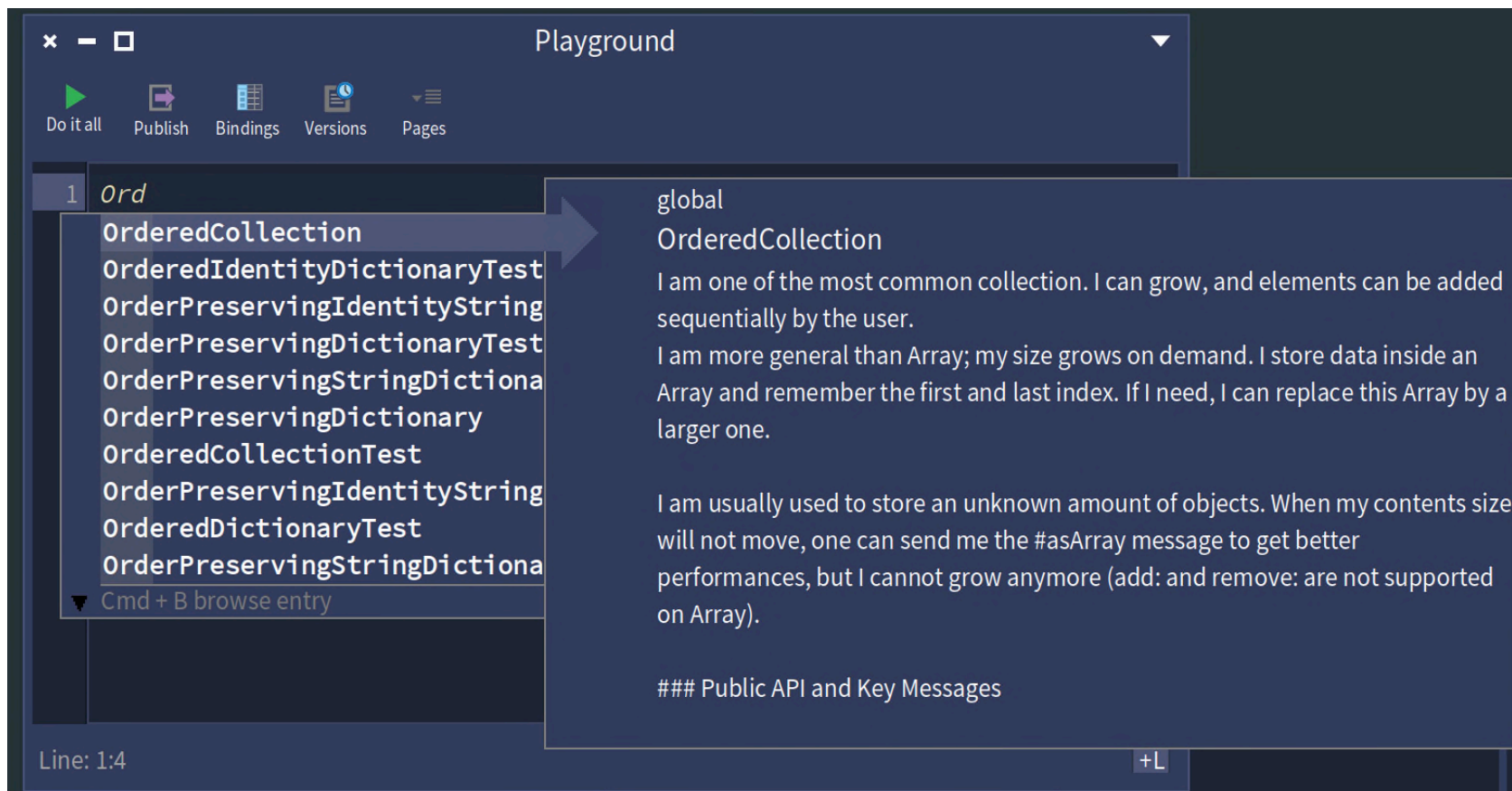
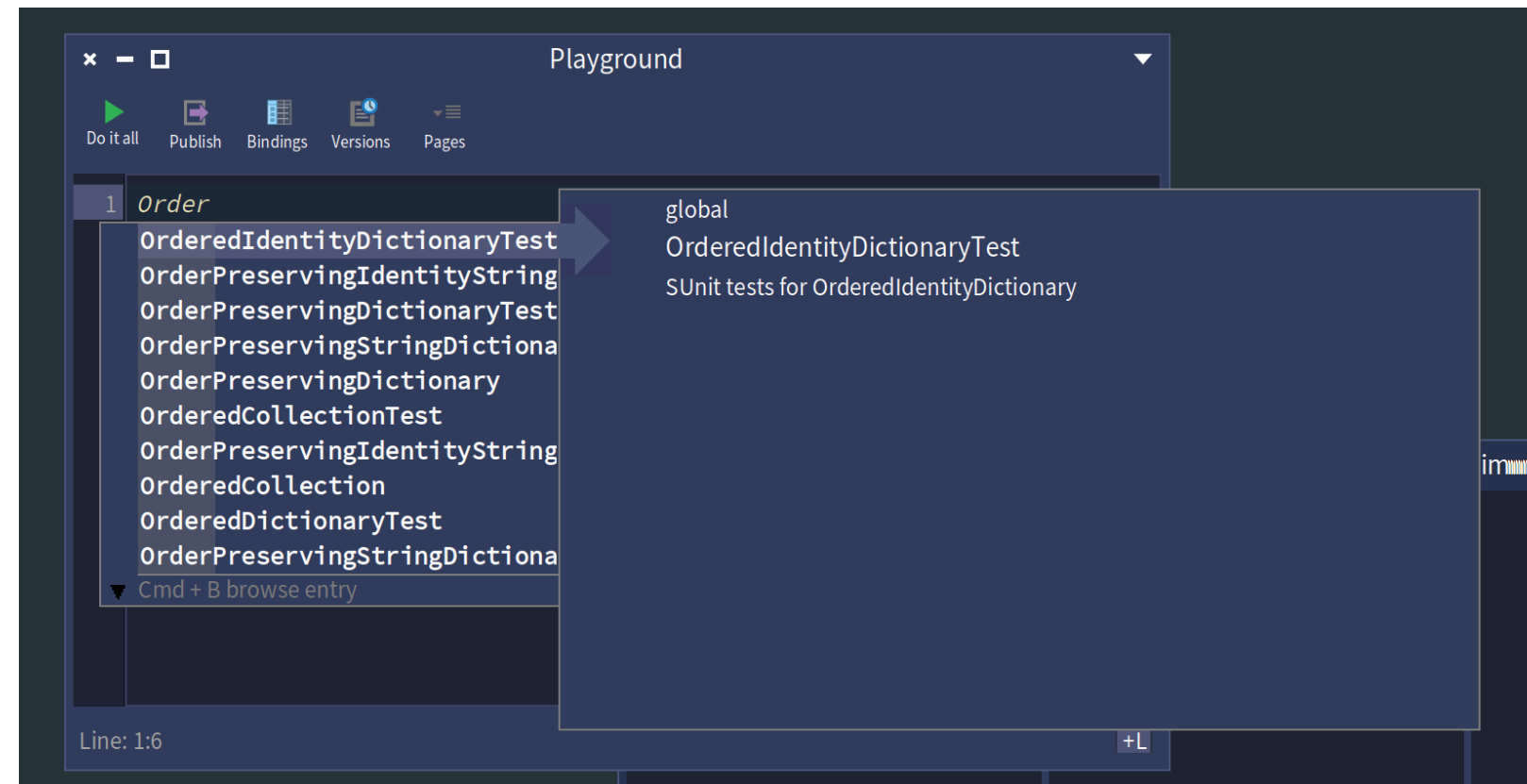
I work mainly as a Dictionary except that I also store the keys in an Array that keeps the order of elements.

I should be used ONLY if you need to keep the keys ordered. Else you should use a Dictionary that is faster and keep less values into memory. (I duplicate the keys). Insertion, update, and inclusion testing have $O(1)$ complexity while removing has $O(n)$ worst-case.

Public API and Key Messages

At the bottom left of the code editor, it says 'Line: 1:5'.

Without With



But with a preemption window

Not have to select OrderedCollection 10 times if you selected OrderedDictionary 10 times before

Favor more recent choices over past ones

Supporting Typos

- Looks simple, but this is quite complex :)
- Ok Edit Distances could be our friends
 - we tried with Levenshtein and Jaro-Winkler
- Impacting the stream architecture of
Complishion

Impact

- “SpPrez” /SpPrezenter”
- so the stream will not propose SpPresenter because it does not match
- Edit distance but they can produce too many noise

Conclusion

- Your efficiency is our concern
- Try the Soft Dev History because this working great!
- Soon “Typo tolerance”